

The banner features a dark blue background with a glowing network of white and light blue lines and nodes, resembling a complex web or data structure. The text "SRv6" is prominently displayed in the upper left, with "SR" in blue and "v6" in green. Below it, "INDUSTRY ROADSHOW" is written in white, and "2026" is in a larger white font. At the bottom right, a tagline in smaller white text reads "INNOVATIONS IN NETWORK PROGRAMMABILITY DRIVING THE FUTURE OF CONNECTIVITY".

SRv6

INDUSTRY ROADSHOW
2026

INNOVATIONS IN NETWORK PROGRAMMABILITY
DRIVING THE FUTURE OF CONNECTIVITY

SRv6 on Hosts

Network and Compute Integration

Abstract

Overview

Integrating Network & Compute

Compute infrastructure is generally disconnected from the IP infrastructure at a protocol level. To produce better price points and allow complete visibility, encapsulation must move directly to the host.

! THE PROBLEM

Protocol-Level Disconnection

Compute remains isolated from IP infrastructure, forcing complex network gear to maintain unsustainable scale and feature loads.

✓ THE SOLUTION

Host & Fabric Integration

Move encapsulation points to the host. This establishes deep network vision and merges the host completely into the IP "fabric."

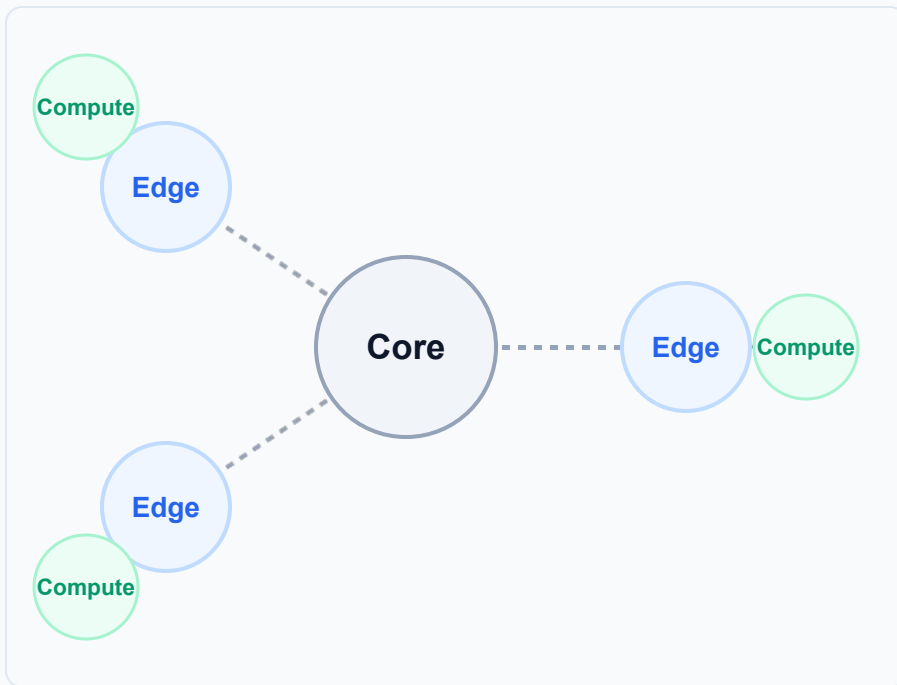
High Level Network Visualization

END-TO-END CHALLENGES

Network Complexity

To achieve end-to-end integration, the infrastructure must overcome multiple layers of fragmentation across domains:

- **3 Domain Types** with distinct boundaries
- **3 Encapsulations** requiring translation
- **3 Configuration Standards** to align
- **3 Traffic Engineering** problems



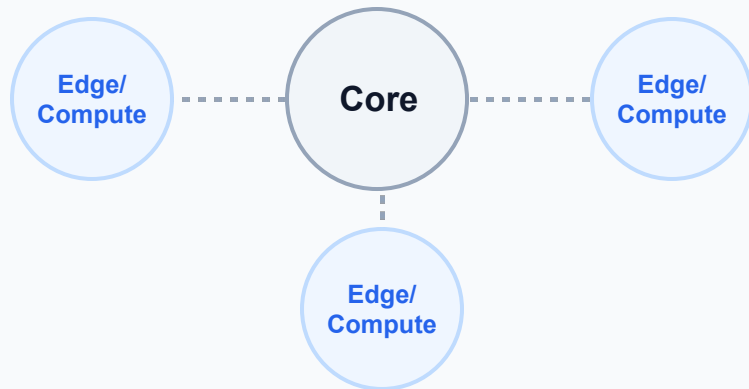
Host/Network Integration

Target

Initial Consolidation

Simplifying the overall network structure through early integration steps:

- **Domain Reduction:** Decreasing the number of distinct boundaries.
- **Encapsulation:** Streamlining encapsulation methods.
- **Standardization:** Moving toward uniform configurations.
- **Traffic Engineering:** Simplifying TE policies within the domain.



SRv6 on Host, Edge and Core

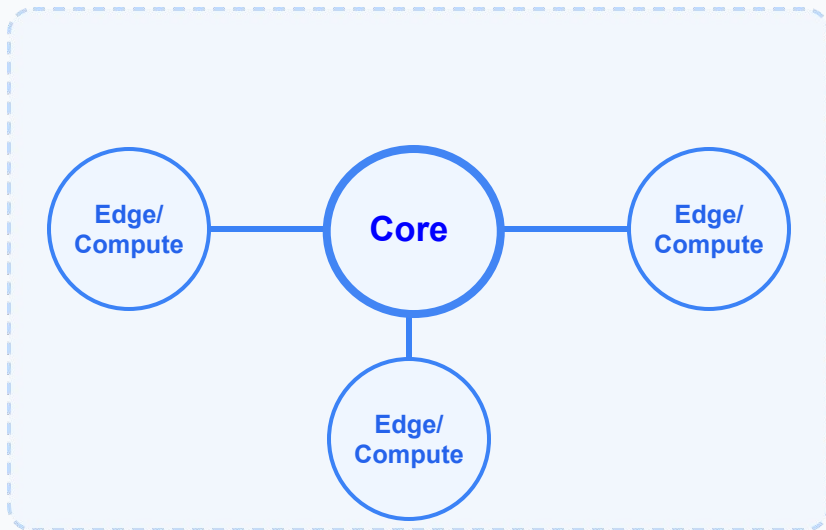
Target

Full Integration

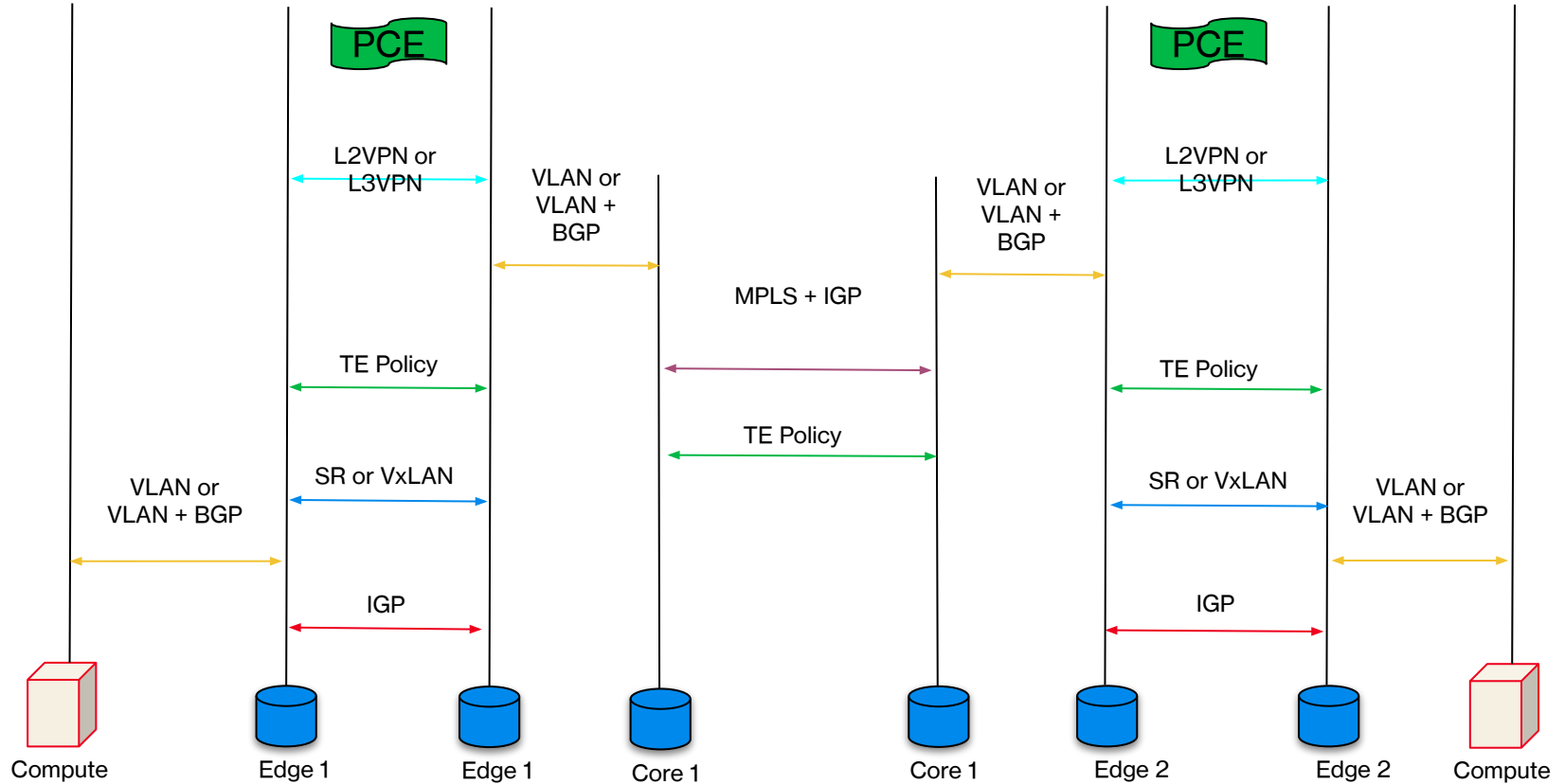
Achieving a completely unified network fabric across all domains with native SRv6:

- **Unified Domains:** SRv6 running natively in all domains.
- **Single Encap:** Eliminating translation with a single encapsulation.
- **One Standard:** Single configuration standard across the network.
- **Single TE Policy:** Unified traffic engineering end-to-end.

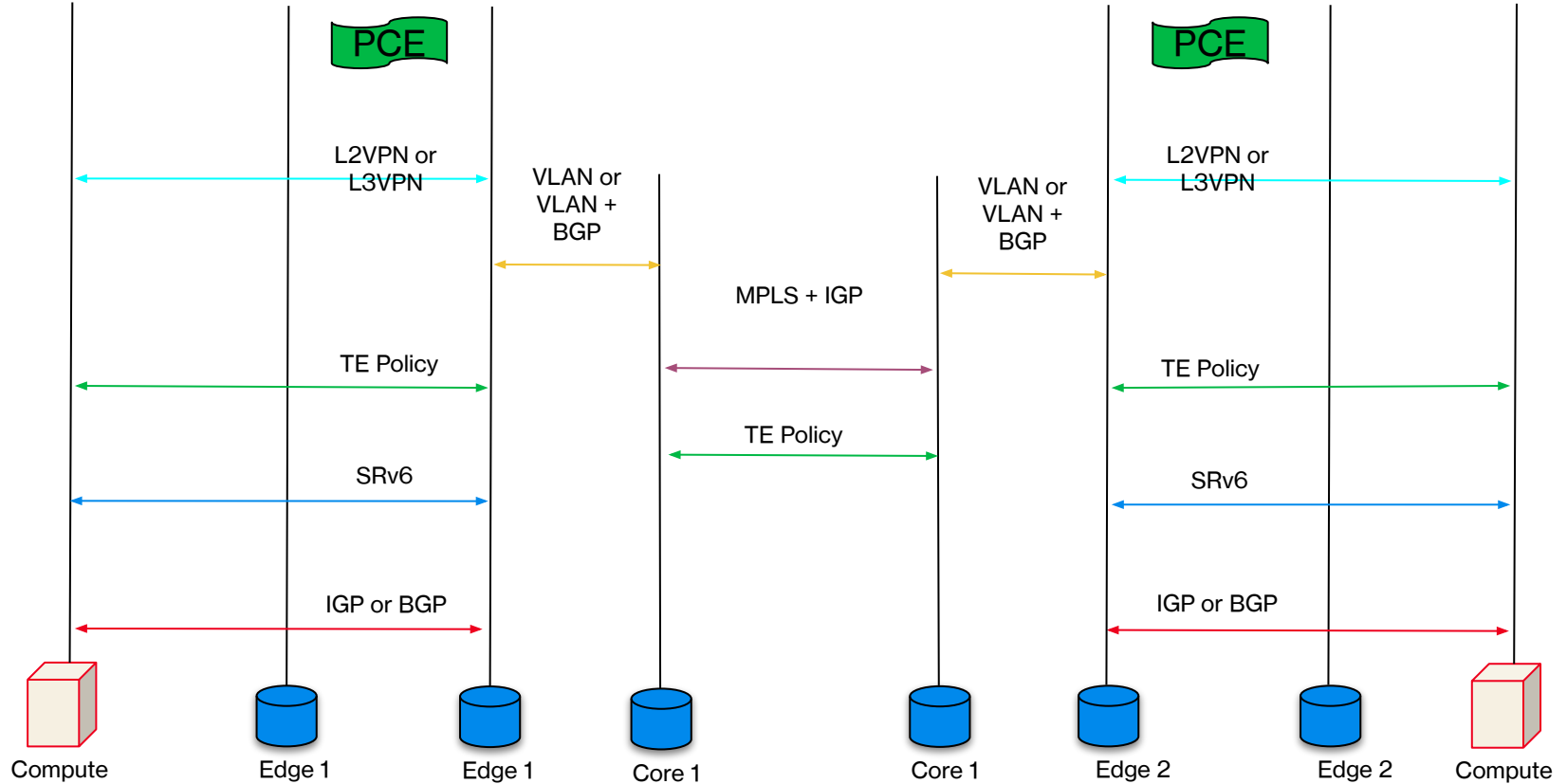
Single SRv6 Encap & TE Policy Domain



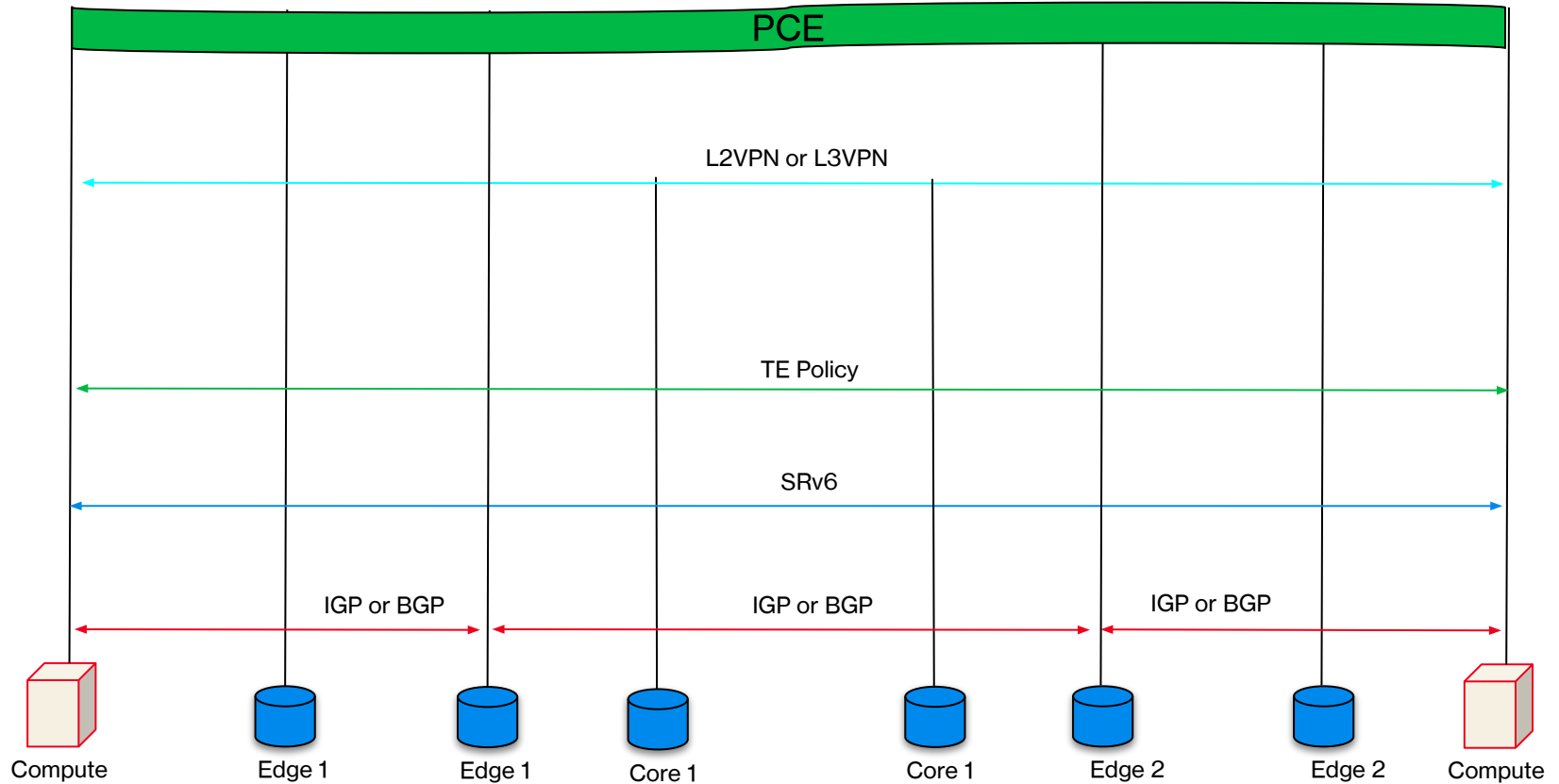
Typical Industry



SRv6 on Host



Design option



Example stack

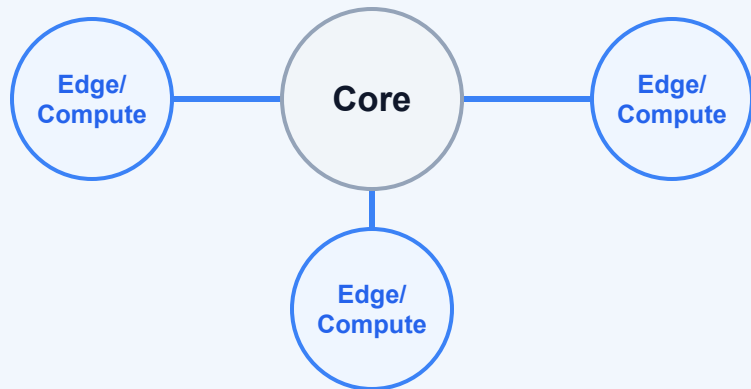
Infrastructure

Network Architecture

The compute must match the network architecture:

- **Internal routing** - IS-IS
- **Abstracted network** - Segment Routing
- **Multi-tenant Overlay** - BGP-EVPN

Single SRv6 Encap & TE Policy Domain



Example stack

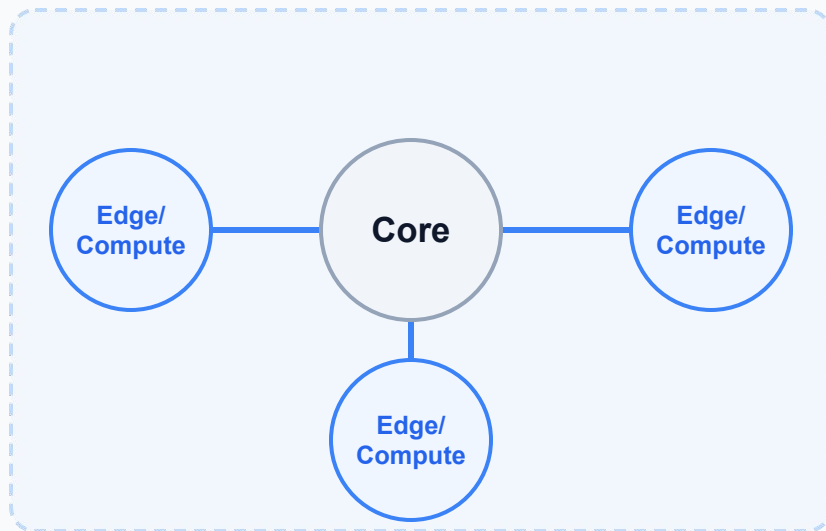
Infrastructure

Compute Architecture

The compute must match the network architecture:

- **Internal routing** - IS-IS
- **Abstracted network** - Segment Routing
- **Multi-tenant Overlay** - BGP-EVPN
- **Virtualization technology** - RHEL Container

Single SRv6 Encap & TE Policy Domain



Proof of Concept

Overview

Integrating Network & Compute

Utilize Free Range Routing(FRR) as the open source control plane teamed with Linux kernel for the data plane.

❗ THE PROBLEM

Protocol-Level Disconnection

FRR lacks several table stakes protocol capabilities to fully integrate with the network.

✅ THE SOLUTION

Host & Fabric Integration

Develop FRR with the requirements, e.g. IS-IS L1/L2, EVPN L2.

Proof of Concept

Overview

Integrating Network & Compute

Engage the FRR community, the Linux community and our other vendors.

⚠ THE PROBLEM

Protocol-Level Disconnection

Changes to the FRR or Linux need to be upstreamed into the community for the additions to remain viable for long-term support.

✅ THE SOLUTION

Host & Fabric Integration

Testing includes full view to the communities and vendors on progression.

Beyond Proof of Concept

Overview

Integrating Network & Compute

There are many applications that vary in requirements across the provider footprints.

⚠ THE PROBLEM

Protocol-Level Disconnection

FRR + Linux may not meet all high speed application requirements.

✅ THE SOLUTION

Host & Fabric Integration

Investigate acceleration/scale options for the applications that require it.